

Planning Support Systems: A New Perspective on Computer-Aided Planning

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This article suggests that the evolving perceptions of planning together with the evolution of computer-based information and techniques provide the foundations for a new perspective on computer-aided planning. It argues first that planners' continued failure to use computers extensively for core planning functions results less from the limitations of hardware and software than from a limited understanding of the proper role these tools should play in planning. It then suggests that the prevailing view of planning has evolved from an image of *planning as applied science* in the 1960s, to *planning as politics* in the 1970s, and then to *planning as communication* in the 1980s. Similarly, it argues that the primary concern of the information sciences has changed from *data* and electronic data processing (EDP) in the 1960s, to *information* and management information systems (MIS) in the 1970s, and then to *knowledge* and decision support systems (DSS) in the 1980s. The article then suggests that the increasingly popular topic of planning support systems (PSS) is best seen as continuing these trends to a broader concern with *intelligence* and collective design. It concludes by briefly considering the implications which this view has for PSS design and implementation.

■ AN ALWAYS IMMINENT REVOLUTION

Computer use in planning began in an optimistic era of steadily increasing real incomes, a widespread faith in the efficacy of science and technology, and the emergence of new academic fields such as operations research, urban economics, and regional science. The general optimism of the 1960s was reflected in professional practice by ambitious, federally funded attempts to develop large-scale metropolitan land use and transportation models and integrated municipal information systems (Hemmens 1971; Government Data Systems 1971; Kraemer 1974; Putman 1979). In planning academia, the optimism was revealed by the popularity of the systems view of planning and the rejection of earlier views of *planning as design* for new models of *planning as an applied science*. For some, the computer offered "a tool of such revolutionary new potential that it may have an effect of redefining the process of planning itself" (Harris 1968a, 223).

This optimism was severely tested in the 1970s with the spectacular failure of the large-scale urban models, the early municipal information systems, and of all attempts to develop policy-relevant, computer-based models (Brewer 1973; Danziger 1977; Lee 1973; Greenberger et al. 1976). Sophisticated analytic tools such as mathematical programming, once assumed to provide the foundations for a new *science of planning*, were found to be increasingly inappropriate for public planning applications. These critiques reflected a more fundamental attack on the "rational planning model" generally assumed to underlie all attempts to use computers in planning.

ABSTRACT

This article examines the changing views of planning and computer-based information that provide the foundations for a new perspective on computer-assisted planning. It begins by tracing the evolving view of planning as *applied science* in the 1960s, as *politics* in the 1970s, and then as *communication* in the 1980s. It then reviews the evolving concern of the information sciences with *data* in the 1960s, *information* in the 1970s, and *knowledge* in the 1980s. It concludes by suggesting that the increasingly popular topic of planning support systems (PSS) can be seen as continuing these trends to include broader concern with *intelligence* and collective design.

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Planners' faith in computer technology was reborn in the 1980s with the development of microcomputers that made advanced information technologies and computer-based methods readily available to professional and academic planners. Microcomputers are now found on nearly every planner's desk; courses in "computer applications" are a standard part of planning curricula; and a wealth of literature on microcomputer applications in planning has appeared (Klosterman 1992, 251-253). These developments have again led technological optimists to ask whether "planning can survive much longer in its present form" (Openshaw 1986, 68).

The 1990s have witnessed yet another "revolution in information technology" resulting from the general availability of geographic information systems (GIS). Planning schools are now rushing to offer GIS courses to satisfy overwhelming student demand; the academic and professional literature is filled with GIS articles; planning conferences are dominated by GIS sessions; and some authors claim that "[v]irtually anything a planner does can be done with GIS" (Robert Lima, quoted in Juhl 1994, 8).

In spite of this continued optimism, it is clear that the oft-foretold revolution in computer-aided planning has yet to occur. Most planners now have microcomputers on their desks, but the available evidence suggests that their usage continues to be very broad, but shallow. That is, while many planners use computers, they are still being used primarily for general-purpose office functions such as processing documents, monitoring budgets, and maintaining records. Computers have proven to be very useful in these applications for increasing office efficiency and facilitating internal administration. However, they continue to find only limited applications in the areas of professional practice that are unique to planning such as forecasting, analysis, and evaluation (Klosterman 1992, 251).

Even in the area of GIS, the available information suggests that this increasingly popular technology is still being used primarily for routine operational and management tasks such as permit processing, collecting and maintaining land-related information, and preparing thematic maps. There is little evidence that GIS is being widely used for planning analysis or evaluation (Wiggins and French 1990; Wiggins 1993). The reasons for this will be examined more fully below.

■ RETURN TO FUNDAMENTALS

Over the last three decades, expensive, fragile, remote, and hard-to-use mainframe computers have been replaced by small, inexpensive, and easy-to-use microcomputers that get faster, cheaper, more powerful, and easier to use every year. The development of fast, high-resolution, full-color display and output devices have spawned spectacular advances in computer-aided design and drafting (CADD), GIS, and visualization software. Equally dramatic improve-

ments in the capacity and speed of data storage and communication devices, from local area networks (LANs) to the Internet, have made a wealth of digital information readily available to practicing planners in the smallest planning agency and firm. Perhaps most important, entirely new kinds of software—most notably electronic spreadsheets, sophisticated database management systems, and graphically oriented multi-tasking operating systems—have made possible a flexible and interactive style of computer interaction unimaginable only ten years ago.

Planners' continued failure to use computers more fully in their professional practice in the face of these rapid—and accelerating—developments suggest that the most important constraints to computer-aided planning may lie in the "soft" side of the technology package: technique, organization, and knowledge (Danziger et al. 1982, 3-5; Masser and Campbell 1991). Some authors suggest the problem lies in the "wicked" nature of planning problems and the fact that local government provide spatially distributed public goods and services that are difficult to describe concretely for computer processing (Dueker 1982, 61; Rittel and Webber 1973). Others point to the mismatch between the structured outputs produced by traditional planning models and decision makers' less formal and unstructured information needs (Langendorf 1985, 422-424). Others highlight the diversity of planners' analytic needs and application areas in a small and poorly funded public-sector market, which provides little incentive for developing computer tools that serve the needs of planners (Klosterman and Landis 1988, 365). Still others point to the planners' isolation from the local government information systems developed primarily to serve operational and administrative needs (Arbeit 1993; Worral 1987).

More fundamentally, recognizing that technology (in the narrow sense of computer equipment) is not the problem suggests that the search for an appropriate role for computer-based information and methods in planning must begin not with a particular technology (or set of technologies) but rather with a conception of planning. This perspective is based on the often-repeated observation, "If the only tool you have is a hammer, everything looks like a nail." This implies in this context that starting with a particular technology such as GIS or the World Wide Web and then seeing how this tool can be applied to planning threatens to distort the nature of planning. It does this almost inevitably—and unconsciously—by focusing attention on those aspects of planning for which a particular tool is appropriate and neglecting other aspects for which it is less appropriate.

Starting with a particular technology also implies that the role of advanced information technologies in planning is shaped largely by developments outside of planning (primarily business and the home entertainment industry) and not by the particular needs of planning. Starting with a

System Optimization	"Planning as applied science"	Information technology is viewed as providing the information needed for a value-neutral and politically neutral process of "rational" planning.
Politics	"Planning as politics"	Information technology is seen as inherently political, reinforcing existing structures of influence, hiding fundamental political choices, and transforming the policy-making process.
Discourse	"Planning as communication"	Information technology and the content of planners' technical analyses are seen as often less important than the ways in which planners transmit this information to others.
Collective Design	"Planning as reasoning together"	Information technology is seen as providing the information infrastructure that facilitates social interaction, interpersonal communication, and debate that attempts to achieve collective goals and deal with common concerns.

Table 1. Evolving views of planning and information technology.

particular technology also does not guide the search for new technologies or help identify ways in which new and existing tools can best be integrated into planning practice. Instead, it takes current technology largely as given and molds planning to fit the technology.

These observations are particularly relevant to the current fascination with GIS among planning academics and practitioners. The popularity of GIS—both within and outside of planning—is due largely to the fact that it is a "chameleon technology" that provides basic functions for collecting, maintaining, analyzing, and displaying spatially related information, functions that are extremely useful for a wide range of public- and private-sector applications. However, this general applicability comes at the expense of particular capabilities that are required for specialized applications such as planning (Harris and Batty 1993, 185). The particular advantages—and limitations—that GIS provides for planning will be discussed briefly below.

These considerations further suggest that while no single technology alone can provide all of the capabilities planners need to carry out their professional responsibilities, a range of different technologies might be used together to provide

capabilities that no single tool can provide on its own. Recognizing this, several authors have adopted Britton Harris's (1989) concept of PSS as an appropriate model for combining a range of computer-based methods and models into an integrated system that can support the planning function (Harris and Batty 1993; Borger and Timmermans 1991; Holmberg 1994; Janowski and Richard 1994; Rugg 1992; Scholten and Padding 1990; Taylor 1991).

Although the concept of PSS has been defined in several ways in the literature, it will be initially defined here as broadly as possible to include all current and future technologies useful for planning. This does not mean a PSS will include all computer-based tools and methods *used by planners*; this definition excludes general-purpose software tools for word processing, report presentation, internal budgeting, and the like that planners use just like other professionals. Instead, it includes only the computer hardware, software, and related information that are *used for planning*, i.e., the information technologies planners use to perform their unique professional responsibilities as planners.¹ Identifying these responsibilities and refining this preliminary definition of PSS will be a primary concern of the remainder of this paper.

■ EVOLVING VIEWS OF COMPUTER-AIDED PLANNING

Suggesting that attempts to define the appropriate role for information technology must start with a conception of planning leads directly to the perennial question: "What is planning?" This is a fundamental (and perhaps unresolvable) issue that cannot be examined in detail here. Instead of addressing this issue directly, the following discussion briefly reviews planners' evolving views of planning and its relation to information technology and computer-based methods. The concluding section of the paper draws on these developments to suggest a useful definition of planning and of planning support systems.

Planning as Applied Science

The introduction of computers into planning in the 1950s and 1960s was part of a more fundamental transition from the profession's traditional concern with the design of the physical city to a new focus on the quantitative techniques and theories of the social sciences. Guided by Mannheim's vision of planning as a rational process that applied scientific knowledge and techniques to the manage-

ment of public affairs, planners abandoned the advocacy of professional visions of the desirable future for the ideology—if not the reality—of value-neutral policy evaluation and the metaphors of the market and pluralist political science. With this, the dominant perception of planning changed from an image of planning as design to planning as an applied science.

This paradigm shift was reflected in planning education by the addition of new students, faculties, and departments from outside the traditional schools of design, and by the abandonment of the once ubiquitous design studio for the now equally prevalent courses in research design, statistics, and quantitative techniques. It was revealed most dramatically in the world of professional practice by the popularity of computer-based urban transportation models, integrated municipal information systems, and large-scale urban models.

The applied science model defined rationality in instrumental terms as finding the best means (actions or policies) for achieving desired ends and planning as an iterative process of defining problems, identifying goals, generating alternatives, and evaluating these alternatives with respect to designated goals. As suggested in Table 1, computers were assumed to play an important role in this task by collecting and storing the required data, providing systems models that could describe the present and project the future, and helping unambiguously identify the best plan from the range of available alternatives (Harris and Batty 1993, 187).²

Underlying this image was an implicit general systems theory that viewed cities as complex dynamic systems made up of hierarchical subsystems and planning as a rational procedure for optimizing overall system properties such as utility or welfare. The knowledge required to simulate urban system functions was assumed to be provided by the new fields of urban economics and regional science. Techniques for selecting the optimal action or policy from the available alternatives were likewise assumed to be derived from then-new analytic procedures such as operations research and linear programming that had proven so effective in military and space applications.

This applied science model of planning was assumed to be equally appropriate for individuals (the planner), private organizations, and public agencies (the planning agency). It was further assumed that: (1) information is a value-free and politically neutral resource; (2) more information is always better; (3) the planner's most important role is providing more and better information that can inform and improve the policy-making process; and, more fundamentally, that (4) a clear distinction could be made between the "objective" facts stored in a computer and the "subjective" opinions and values of individuals and groups. All of these assumptions were severely criticized during the 1970s with the emergence of a new conception of "planning as politics."

Planning as Politics

Planners discovered in the 1970s that public-policy making is fundamentally different from personal or corporate decision making. In the private sector, overall goals and objectives such as maximizing profits or increasing market share are reasonably well defined; reliable information on available alternatives can be collected with little difficulty; and the resources required to achieve designated goals are centrally controlled. None of these conditions apply in the public sector, making it extremely difficult to define—let alone achieve—collective goals and objectives (Rittel and Webber 1973).

Planners also recognized that planning is not—and cannot be—value free. With this came a realization that computer-based information and analytic techniques are themselves inherently and inevitably political, reinforcing existing structures of influence, increasing the power of administrators and technical experts, and transforming the nature of planning and policy analysis. The sheen of objective authority imparted by the neutral technical language was found to inevitably hide inherently political choices in the selection of data and the analysis, presentation, and distribution of results, hiding fundamental political choices within supposedly technical analyses unfathomable to outsiders (Klosterman 1987; Dutton and Kraemer 1979).

The second assumption on the proper role of information in planning was likewise found to be much less straightforward than planners had once assumed. Information needs and the appropriate strategies for dealing with this information have been found to be defined not by public policy issues *per se* but rather by prevailing conceptions of policy goals and of the available means for achieving these goals (Cartwright 1973; Christensen 1985; George 1994).

The seemingly straightforward conversion of information from paper records to digital form can itself have important political and legal implications. Data stored in digital form in a GIS can be used to generate information, such as the location of all land parcels within a kilometer of a hazardous waste site, that would be extremely difficult to obtain when the same data are recorded in handwritten entries stored in file cabinets. As a result, the greatly enhanced ability to store and access information that GIS provides increases the potential for information abuse and misuse, raising fundamental issues of data security, reliability, and responsibility (Aronoff 1989, 269-277).

The introduction of more accurate and more current information can also *increase* public policy conflicts by revealing issues previously hidden from view and fueling policy debates based on conflicting interests and values. Computer-based information systems are of course extremely useful for revealing previously unrecognized inaccuracies and conflicts in existing paper-based property

Data	"Observations are cleaned, coded, and stored in machine-readable form."	Primary concern of electronic data processing (EDP) which promoted efficient transaction processing to improve operational tasks
Information	"Data are organized, analyzed, and summarized into a meaningful form."	Primary concern of management information systems (MIS), which integrated diverse data sets to serve management needs
Knowledge	"Understanding based on information, experience, and study."	Primary concern of decision support systems (DSS), which facilitated semi-structured decision making to support executive decision making
Intelligence	"Ability to deal with novel situations and new problems, to apply knowledge acquired from experience, and to use the power of reasoning effectively as a guide to behavior."	Possible concern of planning support systems (PSS), which will promote discourse and interaction to facilitate collective design

Table 2. *Evolving concerns of information technology.*

records. However, they provide little help in correcting these problems, because errors are often embedded in legal documents that cannot be altered on technical grounds, without some kind of legal procedure (Obermeyer 1994; Werle 1984, 199-202).

Experience has also demonstrated that the social sciences have failed to provide the rigorously tested and empirically confirmed knowledge required to guide the policy-making process (Bernstein 1976, 24-44; Keyfitz 1982; Rein 1976, 52-74). Optimization techniques such as linear programming have been found to be useful for well-defined, narrowly technical problems but largely inappropriate for most planning and public policy issues (Strauch 1974; Harris 1989, 88-89). Underlying these largely practical concerns was a more general attack on the rational planning model that dominated planning theory throughout the decade of the 1970s.

Planning as Communication

The realization that planning was political was enriched during the 1980s by a new generation of "ethnographic" studies of planning practice. These studies revealed that planning involves much more than the collection and provision of information that can (presumably) improve the policy-making process. Planners of course prepare plans and

do analyses. However, they also negotiate, explain, and argue about planning rules, changes, and permissions and administer rules and regulations. Quantitative analysis and related information technologies play an important role in these activities. But so does advice-giving, story-telling, and the other metaphors and rhetorical devices planners use to communicate their ideas to others (Forester 1980; Harris 1989, 86-87; Innes 1990; Krieger 1981; Mandelbaum 1991).

These studies suggest that planning cannot be merely a politically neutral instrumental means for achieving designated ends. Instead, it is an inherently political and social process of interaction, communication, and social design. On this perspective, the *way* in which planners transmit information may be more important than *what* they say. For example, quantitative analyses, no matter how well meaning and technically correct, expressed in bureaucratic terms and technical jargon all too often separate planners from the planned-for, reduce public access to information, increase

the public's dependency on technical experts, and minimize planners' ability to learn from the public (Forester 1989). These problems are only increased as planners' technical tools become more sophisticated and their language becomes more esoteric.

This new *communications* view of planning suggests that it should not be viewed primarily as an abstract decision process attempting to optimize overall system goals such as the community welfare. Instead, it must be seen as including an interactive, open, and ongoing process of inter-subjective communication and collective design in which planners help the relevant community "make sense together while living differently" (Healey 1992). On this view the planning activity is grounded in two realms of rationality: (1) the "collective common sense" and collaborative attempts of groups "deciding and acting" together and (2) the formalized knowledge and the self-reflective consciousness of autonomous individuals (Alexander 1988).

The emergence of the communications view of planning is supported by related developments in the broader realms of social and ethical theory. Philosophers now recognize that while social facts and values are culturally defined, they can be defended and criticized in ways that parallel the methods of science. With this have come new attempts to broaden the narrow positivist view of reason based on scientifically

constructed empirical knowledge to include all realms of discourse. On this view, rationality is based not on pure logic and the abstract evaluation of evidence but rather on an informed consensus formed by a community of individuals in a particular place and time (Brown 1988; Healey 1992, 150-52; Fischer 1990, 217-63; Klosterman 1978; Klosterman 1983).

■ EVOLVING CONCEPTIONS OF INFORMATION SYSTEMS TECHNOLOGY

These developments in the field of planning have been accompanied by equally fundamental but largely independent changes in the prevailing views of the proper role of information technology in public- and private-sector organizations. As shown in Table 2, these developments can be viewed broadly as an evolving concern with: (1) *data* during the 1960s, (2) *information* during the 1970s, and then (3) *knowledge* in the 1980s.³

EDP and Data

In the 1960s the infant computer industry was concerned mainly with the handling of *data*, direct observations which have been cleaned, coded, and stored in machine-readable form.⁴ Computer-based information systems were devoted almost exclusively to electronic data processing (EDP), the automation of existing manual procedures, and the computerization of vast amounts of routine data previously stored on paper. The goal was to improve routine operations by processing transactions more efficiently, integrating data for related functions, and generating tabular summary reports (computer printouts) useful for improving day-to-day operations. The prevailing technology was the batch processing of custom-designed, single-purpose, transaction-based information systems on mainframe computers. Public-sector applications included the computerization of day-to-day operational functions such as payroll, accounting, and property tax assessment (Huxhold 1991, 5-9).

■ MIS AND INFORMATION

In the 1970s the emphasis changed from data to *information*, i.e., data that have been organized, analyzed, and summarized into a meaningful form. The primary concern shifted from the efficient processing of data for operational needs to management information systems (MIS) and the structuring and synthesis of these data in forms that could serve management needs. The principle systems objectives became the integration of EDP tasks by organizational function, the processing of information queries, and the generation of summary reports based on a comprehensive database. At the same time, the technology evolved to include minicomputers, relational database management systems (DBMS), and computer-based mapping. Public sector applications included the development of urban information

systems (UIS), GIS, and land information systems (LIS) that together provided the integrated information required for management functions such as permit processing, code enforcement, infrastructure management, and transit operations. These management functions continue to be the primary use of most urban GIS/LIS systems (Grimshaw 1988; Huxhold 1991, 64-126).

DSS and Knowledge

The emphasis changed again in the 1980s from information to *knowledge*, i.e., to understanding based on information, experience, and study. The change in emphasis reflected decision makers' desire for analytical modeling capabilities and for more interaction with solution processes than were provided by management information systems of the 1970s. Stimulated by Gorry and Morton's seminal paper, an entirely new type of system, decision support systems, emerged, generating a substantial body of theory and a large number of applications (Gorry and Morton 1971; Keen and Morton 1978; Sprague and Carlson 1982; Sprague and Watson 1986).

Two primary features characterize the DSS of the 1980s. First, the systems are composed of three components: a user interface, a database, and a model base. Together, these three components are designed to provide a framework that integrates (1) all relevant information from a wide variety of sources with (2) the full range of analytical and statistical modeling tools and (3) a graphical interface that conveys information in a readily understandable form to decision makers. The primary design objective is providing flexibility and quick response in an easy-to-use system that could be adapted to the evolving needs and styles of system users.

More importantly, DSS differs from previous MIS approaches in being deliberately designed to deal with ill- or semi-structured decisions. For well-structured problems, the fundamental variables of concern and the relationship between these variables are easy to identify and measure and, thus, to represent in a mathematical or computer model. In contrast, poorly structured problems can be hard to measure and partially qualitative in nature, requiring value judgments, in part because decision makers are unable—or unwilling—to articulate clearly the objectives of the analysis or the characteristics of desired solutions (Gorry and Morton 1971; Langendorf 1985, 424).

DSS is designed to help deal with poorly structured decisions by facilitating iterative, integrative, and participative decision processes. They are iterative because they allow decision makers to systematically generate and evaluate a number of alternative solutions, gaining insights that are introduced as inputs to help guide further analyses. They are integrative in incorporating decision makers' substantive knowledge along with the quantitative data provided by the models to evaluate alternatives across a broad range of pertinent criteria. And they are participatory by permitting

decision makers to examine the consequences of employing different information and modeling approaches and to choose alternate decision criteria, objectives, and constraints.

The principles underlying the DSS model have been revealed most clearly in the public sector by the spatial decision support systems (SDSS) developed by Paul Densham and others (e.g., Armstrong and Densham 1990; Densham and Rushton 1988). SDSS is designed to support decision making related to complex spatial problems such as determining the optimal location for service centers such as elementary schools and fire stations. As the name implies, the system incorporates three spatially related components of a traditional DSS: a database containing a variety of spatial and nonspatial data, spatially related analytic and simulation models, and a user interface that generally includes a GIS. The GIS generates and stores the required spatial data that are used as input data for the analytic models. The models take the required data from the database, use them to conduct the required analysis or simulation, and store model outputs in the database. GIS and other data visualization software are then used to display the model results, generally in map form.

■ TOWARD A PLANNING SUPPORT SYSTEM

The parallel (if largely unrelated) developments in the spheres of planning and information systems technology together provide the foundations for a richer conception of the role advanced information technologies can play in planning. In planning, earlier views of planning as an objective and value-neutral tool have been replaced by an increasingly sophisticated appreciation of the inherently political nature of the planning process. With this has come the realization that planning is best seen as including not only the activity of an isolated individual or organization but also—and perhaps more importantly—as an ongoing process of social design, interactive dialogue, and debate in which planners, public officials, and the public attempt to decide together how to best manage the collective concerns of society (Healey 1992). In the realm of information technology, the evolving concern with operational data, management information, and then with executive-level knowledge has moved the perceived role of information technology further up the organizational pyramid and from routine, well-structured concerns to increasingly less-routine, less well-structured, and much more difficult issues (Huxhold 1991, 12-22).

As shown in Table 2, these developments suggest a possible next evolutionary stage beyond a concern with supplying the knowledge to support executive decisionmaking at the peak of the organizational pyramid. At the next stage, planners may attempt to provide *intelligence*—the ability to deal with novel situations and new problems, to apply knowledge acquired from experience, and to use the power of reasoning effectively as a guide to behavior. This approach would expand the role of information technology beyond the organization to address inter-organizational issues of information sharing and mutual

learning. It suggests that planning support systems should facilitate *collective design*—social interaction, interpersonal communication, and community debate that attempts to achieve collective goals and deal with common concerns.

This is of course an extremely idealized and abstract view of what planning support systems are and should become. However, it does suggest that PSS, like current DSS (and SDSS) should be designed to provide interactive, integrative, and participatory procedures for dealing with non-routine, poorly structured decisions. However, as a *planning* system, it must also pay particular attention to long-range problems and strategic issues and explicitly facilitate group interaction and discussion; this is in contrast to DSS and SDSS, which are generally designed to support shorter-term policy making by isolated individuals and organizations.

It also suggests PSS must not be seen as a radically new form of technology that will replace the software tools planners currently find on their desks. Instead, it must take the form of an *information framework* that integrates the full range of current (and future) information technologies useful for planning. This perspective recognizes that the design of a *planning* support system must start with the application—planning—and not with a particular technology (or technologies) and should use all available tools appropriate for serving the particular needs of planning. It also recognizes that most planning problems are composed of different problem areas, which often have their own suitable technologies, and any comprehensive PSS must utilize a full range of different technologies, each appropriate for a particular type of function (Han and Kim 1989, 301).

PSS should also not be viewed as a closed black-box collection of computer models that can accept raw data and automatically generate master plans, forecast future land use patterns, or identify optimal policies and actions. Instead, it must be seen as providing the *information infrastructure* for planning that facilitates interaction among planners, and between planners and other actors, both within and outside of government. The PSS must contain structured and accessible information (not just raw data) about the real world features of concern to the public, along with a full range of software tools for analysis, forecasting, and decision making. It must also support a continuous and interactive process of analysis, design, and evaluation that constantly integrates new information generated as model-produced analytical results redefine design issues and the elaboration of design issues generates new demands for analytical information (Harris and Batty 1993, 189; Rugg 1992, 223; Schuur 1994, 102).

■ THE ROLE OF GIS

The heart of any PSS will undoubtedly be a GIS. The GIS will serve first as a display and communication device, producing maps and charts that describe past and present conditions and model outputs that suggest alternative futures. However, its spatial analysis capabilities will also

play an essential role in generating new spatially referenced information required by the computational components of the system.

However, a PSS cannot consist of a GIS alone. It must also include the full range of planners' traditional tools for urban and regional economic and demographic analysis and forecasting, environmental modeling, transportation planning, and predicting future development and land use patterns. It must also include other technologies such as expert systems (Han and Kim 1989), decision support aids such as multi-attribute utility theory (Lee and Hopkins 1995), hypermedia systems (Shiffer 1992), and group decision support systems (Armstrong 1993; Finaly and Marples 1992).

GIS alone cannot serve all the needs of planning because the current generation of general purpose GIS cannot easily accommodate the particular information, computational, and display needs of planning. Among other things, planning requires: (1) information that is effectively "aspatial" at a particular level of analysis (e.g., regional population and employment levels and trends for the analysis of a city and its components); (2) information over time (e.g., population, employment, and land use data for the past, present, and possible futures); and (3) measures of spatial interaction (e.g., the number of trips between zones). None of these can be easily incorporated into current GIS (Harris and Batty 1993, 190).

Planning analysis and forecasting also require computational procedures that go beyond the standard database manipulation and Boolean functions of most GIS. These needs include the ability to: (1) compute cross products, manipulate matrices, and conduct statistical analyses; (2) estimate system parameters and derive solutions iteratively; and (3) support analytical and design functions that incorporate goals, objectives, costs, and benefits (Couclelis 1991; Harris and Batty 1993, 189). Until recently, GIS has also been unable to produce basic data displays such as bar graph and scatter charts, nor could it integrate maps with graphic images, sound and video clips, or other display media. The software still is unable to produce specialized graphic displays such as population pyramids that are particularly useful for planning.⁵

As a result, it is highly unlikely that all of planners' computational needs will ever be incorporated into standard GIS packages. Instead, planners will have to adapt existing GIS tools to meet their needs. Traditional programming languages can be used to develop spatial analysis and modeling tools entirely independent of commercial GIS packages. A combination of sophisticated GIS "macro" commands and traditional programming languages can also be used to develop analytic models closely linked to full-featured GIS "toolkits" such as ARC/INFO (Batty and Xie 1994a; Batty and Xie 1994b). Alternatively, GIS can be used to prepare maps and perform spatial analyses when

loosely coupled to other software tools, such as electronic spreadsheets, that provide the computational functions GIS does not offer with data transferred as needed from one software platform to another (Ding and Fotheringham 1992; Harris and Batty 1993, 189-191).⁶

■ INCONCLUSION⁷

The ideal PSS is as easy to define as it will be difficult to implement. The perfect PSS would be a fully integrated, flexible, and user-friendly system that allows the user to: (1) select the appropriate analysis or forecasting tool from an "intelligent digital toolbox" that helps the user identify the most appropriate methodologies and tools for dealing with a particular task, (2) link the appropriate analytic or projection model to the required local, regional, or national information stored—or accessed—through the PSS, (3) run the appropriate model(s) to determine the implications of alternative policy choices and different assumptions about the present and the future, and (4) instantaneously view the results graphically in the form of charts, maps, and interactive video/sound displays. The process would ideally be done interactively and in real time, using models and mapped data that are available locally or over the Internet. The system would incorporate both qualitative and quantitative decision aids, as appropriate, facilitating voting, ranking, and group interaction (Finaly and Marples 1992, 99; Harris 1989; Rugg 1992, 225; Holmberg 1994, 11-14; Harris and Batty 1993, 193-94).

Many of the required analytic and display modules have been developed and are available, at least in prototype form, somewhere. The important—and difficult—task is integrating these modules into a coherent system that serves the needs of planners. To do this, standardized formats will have to be developed for data storage, access, and interchange; efforts in this direction have begun such as the development of the federal government's spatial data transfer standard (SDTS). Software tools will have to be identified or developed that can deal with particular tasks, utilizing data stored in standard formats or generated by the analytic modules. The analytic and display models must then be combined into an integrated package and new linking tools must be developed that can transfer data from one software tool to another.

This paper has not attempted to provide a well-structured blueprint to guide these developments, either now or in the future. Issues of systems design and implementation can only be resolved slowly and painfully over time, as planning scholars and practitioners develop the software tools and experience that will make the PSS ideal a reality. In this way, the paper ends inconclusively, leaving unanswered a host of extremely difficult and important theoretical and practical questions of determining the best ways in which the array of available and emerging information technologies can be integrated into a comprehensive system to support

the needs of planning. Crucial issues of funding, coordination, training, and data dissemination are also left untouched.

Nevertheless, this article has attempted to provide a general framework within which these developments can occur and a vision of where they should be directed. The vision begins by recognizing that the search for an appropriate role of information technology in planning must begin not with technology but with planning and must strive to adapt available technologies to the particular needs of planning. Emerging conceptions of planning as collective design suggest that planners' computer-based tools must be directly accessible to the public and address issues that the public is most interested in (e.g., exploring the likely social, fiscal, and environmental impacts of alternative development proposals) in ways they can readily understand.⁸ If this vision is correct, advanced information technologies may finally take their place at the center of professional planning practice, supporting community planning in its fullest and richest forms.

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■ NOTES

1. Defining PSS to include "all of the technologies used by planners" is equivalent to defining planning as "everything that planners do." This definition of planning is too broad because it includes not only the things planners do as planners but also the things they do as private citizens (e.g., their interaction with family and friends) or even in violation of the norms of planning (e.g., taking bribes). It is also too narrow because it excludes the activities of persons who may not be identified as planners but who carry out planning functions (as they may be defined in a particular case). Similar definitional problems plague an overly broad—and narrow—definitions of PSS.
2. This "rational" model still underlies many views of the proper role of computers and related technologies in planning (Webster 1993; Webster 1994; Batty 1995, 27-28).
3. This four-part distinction between data, information, knowledge, and a new stage of intelligence discussed later in the paper is adapted—in a modified form—from similar concepts proposed by Catanese (1979, 91-93).
4. The following discussion draws on similar discussions in Hopple (1988, 6-7) and Mannheim (1987, 122-123).
5. A comprehensive review of more fundamental "post-positivist" critiques of GIS is provided in Sui (1994).
6. These alternative strategies are evaluated in Sui (1994, 258-260).
7. The idea of ending a paper with a section titled "inconclusion" is adapted from Britton Harris.
8. An attempt to provide this kind of collaborative planning support is provided by the *What if?* PSS. For current information on this system, see the *What if?* Web page at <http://www.dcgis.com/whatif/whatif.htm>.

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